a.s.

The Recursive Audit Theorem: Self-Reference of the Audit Sword, Convergence under $\rho < 1$, and the Fixed-Point Identity with Theorem 3

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Abstract

The SCX "Audit Sword" principle asserts that every use of the Cercis Score must be independently auditable. But this sword can and must be turned upon itself. If auditor $\mathcal{A}_1$ issues a noise estimate $\hat{\eta}_1$, a meta-auditor $\mathcal{A}_2$ can audit $\mathcal{A}_1$'s output, producing $\hat{\eta}_2$, and so on, ad infinitum. We formalize this **Recursive Audit Problem** and prove four results. (i) **Recursive Convergence** (Theorem **??**): if each meta-audit layer's additional noise σ_k^2 satisfies $\sigma_k^2 \leq \rho\sigma_{k-1}^2$ with $\rho < 1$ (the meta-auditor is strictly higher-fidelity than its object), the recursive sequence $\{\hat{\eta}_k\}$ converges almost surely to a fixed point η^* satisfying $\eta^* = f(\eta^*)$, with exponential rate when $\rho \leq 1/2$. This is **rigorous** from Banach's fixed-point theorem and Doob's martingale convergence theorem. (ii) **Infinite Regress Inevitability** (Theorem **??**): if $\rho \geq 1$ (meta-auditor no more precise than the object auditor), the probability of divergence tends to 1 the Audit Sword *breaks* under self-reference. [Rigorous] (Pólya's theorem). (iii) **Fixed-Point Information-Theoretic Characterization** (Conjecture **??**): η^* is conjectured to equal the audit heat death $H_{\min} = H(\varepsilon \mid S)$ from AE-Theorem the convergence point of recursive auditing is exactly Theorem 3's information barrier. [Open] (equivalence proof). (iv) **Gödel Analogy** (Open Problem **??**): the $\rho \geq 1$ divergence may correspond to an **absolutely unauditable audit proposition** a statement about audit quality that cannot be verified by any finite iteration of the Audit Sword. [Open]

[Rigorous][Conditionally Rigorous][Open][Honest Critique]

1 Introduction

Self-reference is the crucible in which foundational claims are tested. Gödel's incompleteness theorems showed that any sufficiently powerful formal system contains true statements it cannot prove. Turing's halting problem showed that no algorithm can decide whether an arbitrary program terminates. The SCX framework makes an equally universal claim the Audit Sword: every use of the Cercis Score can be independently audited and must therefore face the same self-referential test.

The recursive audit problem is this: **Can the audit of an audit be audited? And if so, does this recursion converge, or does it spiral into infinite regress?**

Concretely:

(1) Auditor $\mathcal{A}_0$ examines dataset $\mathcal{D}$ and outputs an estimated noise rate $\hat{\eta}_0$ with confidence interval $[\hat{\eta}_0 - \epsilon_0, \hat{\eta}_0 + \epsilon_0]$;

(2) Meta-auditor $\mathcal{A}_1$ examines $\mathcal{A}_0$'s output (and the underlying data) and produces $\hat{\eta}_1$ with confidence $[\hat{\eta}_1 - \epsilon_1, \hat{\eta}_1 + \epsilon_1]$;

(3) Meta-meta-auditor $\mathcal{A}_2$ examines $\mathcal{A}_1$'s output, producing $\hat{\eta}_2$;

(4) $\cdots$

(5) $\mathcal{A}_k$ produces $\hat{\eta}_k$.

The question is whether $\hat{\eta}_k$ converges as $k \to \infty$, and if so, to what value and under what conditions.

Contributions.

(1) **Recursive Convergence** (Theorem **??**): $\rho < 1 \Rightarrow \hat{\eta}_k \xrightarrow{a.s.} \eta^*$. [Rigorous]

(2) **Infinite Regress** (Theorem **??**): $\rho \geq 1 \Rightarrow P(\text{convergence}) \to 0$. [Rigorous]

(3) **Fixed-Point Identity Conjecture** (Conjecture **??**): $\eta^* = H_{\min}$. [Open]

(4) **Gödel Analogy** (Open Problem **??**): absolutely unauditable propositions. [Open]

2 Preliminaries

2.1 SCX

SCX Thm14

- $\{\hat{\eta}_k\}_{k=0}^{\infty}$ $\hat{\eta}_k \in [0,1]$ k
- $f : [0,1] \to [0,1]$
- ρ
- Banach
- Doob L_1
- Pólya
- $H_{\min} = H(\varepsilon \mid S)$ AE-

2.2

Definition 2.1 (). $\mathcal{A}_0$ $\mathcal{D}$ $k \geq 1$ k $\mathcal{A}_k$

(i) $\mathcal{D}$ *Cercis*

(ii) $\{\hat{\eta}_j, \epsilon_j\}_{j=0}^{k-1}$

(iii)

$\mathcal{A}_k$ $\hat{\eta}_k$ ϵ_k $\{\hat{\eta}_k\}_{k=0}^{\infty}$

Definition 2.2 (). $(\Omega, \mathcal{F}, P)$ $\{\mathcal{F}_k\}_{k=0}^{\infty}$

$$\mathcal{F}_k \ := \ \sigma(\hat{\eta}_0, \hat{\eta}_1, \ldots, \hat{\eta}_k),$$

$\mathcal{F}_k$ k σ- $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}$

Definition 2.3 (). ρ $\mathcal{A}_{k+1}$ $\mathcal{A}_k$

$$\sigma_{k+1}^2 \ \leq \ \rho \ \cdot \ \sigma_k^2, \tag{1}$$

$\sigma_k^2 = \mathbb{V}[\hat{\eta}_k \mid \eta^*]$ k η^*

$\rho < 1$ $\rho = 1$ $\rho > 1$

Definition 2.4 (SCX). $f : [0,1] \to [0,1]$

$$\hat{\eta}_{k+1} \ = \ f(\hat{\eta}_k) \ + \ \xi_k, \tag{2}$$

ξ_k $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ $\mathbb{V}[\xi_k \mid \mathcal{F}_k] = \sigma_k^2 f$ k $k+1$

3 Main Results

3.1 Theorem RA.1

3.1 (/ Recursive Audit Convergence).

Formal Statement

Assumption 3.1 (/ Contraction). *SCX* $f : [0,1] \to [0,1]$ $[0,1]$ $\rho_f \in [0,1)$

$$\forall \eta, \eta' \in [0,1] : \quad |f(\eta) - f(\eta')| \;\le\; \rho_f \, |\eta - \eta'|.$$

Assumption 3.2 (/ Noise Decay). $\{\sigma_k^2\}_{k=0}^{\infty}$

$$\sigma_{k+1}^2 \;\le\; \rho_\sigma \, \sigma_k^2, \qquad \rho_\sigma \in [0,1).$$

$\rho := \max(\rho_f, \rho_\sigma) < 1$

Assumption 3.3 (/ Boundedness). $[0,1]$ $P(\hat{\eta}_k \in [0,1]) = 1$ $k \ge 0$ $D_k := \hat{\eta}_k - \eta^*$ $|D_k| \le 1$ *a.s.*

Assumption 3.4 (/ Martingale Difference). ξ_k $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ *a.s.* $\mathbb{E}[\xi_k^2 \mid \mathcal{F}_k] = \sigma_k^2$ *a.s.* σ_k^2 $\mathcal{F}_k$-

Theorem 3.5 (Recursive Audit Convergence /). **??**–**??**

(i) $\hat{\eta}_k \xrightarrow{a.s.} \eta^*$ $k \to \infty$ $\eta^* = f(\eta^*)$ f $[0,1]$

(ii) $\rho \le 1/2$ $C < \infty$ $k \ge 1$

$$|\hat{\eta}_k - \eta^*) \;\le\; C \, \sigma_0 \, \rho^{k/2} \sqrt{\log k}, \tag{3}$$

$1 - 1/k^2$ *Borel–Cantelli*

.

Banach

?? $[0,1]$ $\mathbb{R}$ f Banach

- $\eta^* \in [0,1]$ $\eta^* = f(\eta^*)$
- $\eta_0 \in [0,1]$ $\eta_{k+1} = f(\eta_k)$

$$|\eta_k - \eta^*| \;\le\; \rho_f^k \, |\eta_0 - \eta^*|.$$

$\{D_k\}_{k=0}^{\infty}$ $D_k := \hat\eta_k - \eta^*$ (**??**)

$$D_{k+1} = \hat\eta_{k+1} - \eta^* = f(\hat\eta_k) + \xi_k - \eta^* = f(\eta^* + D_k) - f(\eta^*) + \xi_k.$$

$\eta^* = f(\eta^*)$

$\mathcal{F}_k$ ξ_k **??**$\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ a.s. D_k $\mathcal{F}_k$-

$$\mathbb{E}[|D_{k+1}| \mid \mathcal{F}_k] \;\le\; \mathbb{E}\big[\,|f(\eta^* + D_k) - f(\eta^*)| \;\big|\; \mathcal{F}_k\,\big] \;+\; \mathbb{E}[|\xi_k| \mid \mathcal{F}_k].$$

??

$$|f(\eta^* + D_k) - f(\eta^*)| \;\le\; \rho_f\,|D_k| \quad \text{a.s.}$$

$|D_k|$ $\mathcal{F}_k$- $\rho_f|D_k|$

Jensen **??**

$$\mathbb{E}[|\xi_k| \mid \mathcal{F}_k] \;\le\; \sqrt{\mathbb{E}[\xi_k^2 \mid \mathcal{F}_k]} \;=\; \sigma_k \quad \text{a.s.}$$

??$\sigma_k \le \rho_\sigma^{k/2}\,\sigma_0$

$$\mathbb{E}[|D_{k+1}| \mid \mathcal{F}_k] \;\le\; \rho_f\,|D_k| \;+\; \rho_\sigma^{k/2}\,\sigma_0 \quad \text{a.s.}$$

$\rho := \max(\rho_f, \rho_\sigma) < 1$ $\rho_f \le \rho\rho_\sigma \le \rho$

$$\mathbb{E}[|D_{k+1}| \mid \mathcal{F}_k] \;\le\; \rho\,|D_k| \;+\; \rho^{k/2}\,\sigma_0 \quad \text{a.s.} \tag{4}$$

Doob

$$M_k \;:=\; |D_k| \;+\; \frac{\sigma_0}{1-\sqrt{\rho}}\,\rho^{k/2}.$$

$\{M_k\}$ $\{\mathcal{F}_k\}$

$$\begin{aligned}
\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] &= \mathbb{E}[|D_{k+1}| \mid \mathcal{F}_k] \;+\; \frac{\sigma_0}{1-\sqrt{\rho}}\,\rho^{(k+1)/2} \\
&\le \rho\,|D_k| + \rho^{k/2}\sigma_0 \;+\; \frac{\sigma_0}{1-\sqrt{\rho}}\,\rho^{(k+1)/2} \qquad (\textbf{??}) \\
&= \rho\,|D_k| \;+\; \sigma_0\rho^{k/2}\left(1 + \frac{\sqrt{\rho}}{1-\sqrt{\rho}}\right) \\
&= \rho\,|D_k| \;+\; \sigma_0\rho^{k/2}\,\frac{1}{1-\sqrt{\rho}} \\
&\le \rho\left(|D_k| \;+\; \frac{\sigma_0}{1-\sqrt{\rho}}\,\rho^{k/2}\right) \\
&\le |D_k| \;+\; \frac{\sigma_0}{1-\sqrt{\rho}}\,\rho^{k/2} \;=\; M_k \quad \text{a.s.}
\end{aligned}$$

$\rho < 1$ $\{M_k\}$ $\{\mathcal{F}_k\}$-

Doob

??$|D_k| \le 1$ a.s. M_k L_1

$$\mathbb{E}[|M_k|] \;\le\; 1 + \frac{\sigma_0}{1-\sqrt{\rho}} \;<\; \infty, \quad \forall k.$$

Doob Doob, 1953 L_1 M_∞ $\mathbb{E}[|M_\infty|] < \infty$

$M_k \xrightarrow{a.s.} M_\infty$ $\rho^{k/2} \to 0$ $|D_k| \xrightarrow{a.s.} 0$ $\hat{\eta}_k \xrightarrow{a.s.} \eta^*$ (i)

$\rho \le 1/2$ **Hájek–Rényi**

$\rho \le 1/2$ Hájek–Rényi

D_k

$$D_{k+1} - D_k = f(\hat{\eta}_k) - f(\hat{\eta}_{k-1}) + \xi_k - \xi_{k-1}.$$

$$D_k = f(\hat{\eta}_{k-1}) - \eta^* + \xi_{k-1} = \big(f(\eta^* + D_{k-1}) - f(\eta^*)\big) + \xi_{k-1}.$$

$S_k := \sum_{j=0}^{k-1} A_j$ $A_j := \mathbb{E}[D_{j+1} \mid \mathcal{F}_j] - D_j$ $\Delta_k := D_k - \mathbb{E}[D_k \mid \mathcal{F}_{k-1}]$
Hájek–Rényi

X_k $X_0 = 0$ $k \ge 1$

$$X_k \;:=\; \sum_{j=0}^{k-1} \xi_j.$$

$\mathbb{E}[\xi_j \mid \mathcal{F}_j] = 0 \{X_k, \mathcal{F}_k\}$ quadratic variation

$$\langle X\rangle_k \;=\; \sum_{j=0}^{k-1} \mathbb{E}[\xi_j^2 \mid \mathcal{F}_j] \;=\; \sum_{j=0}^{k-1} \sigma_j^2.$$

??$\sigma_j^2 \le \rho_\sigma^j \, \sigma_0^2 \le \rho^j \, \sigma_0^2$

$$\langle X\rangle_k \;\le\; \sigma_0^2 \sum_{j=0}^{k-1} \rho^j \;=\; \sigma_0^2 \, \frac{1-\rho^k}{1-\rho}.$$

Hájek–Rényi Kolmogorov $\{X_k\}$ $\{c_k\}$

$$P\Big(\sup_{m\ge k} |X_m| > \varepsilon\Big) \;\le\; \frac{1}{\varepsilon^2} \sum_{m=k}^{\infty} \frac{\mathbb{E}[(X_{m+1} - X_m)^2]}{c_m^2}.$$

$c_m = 1$ m $\mathbb{E}[(X_{m+1} - X_m)^2] = \mathbb{E}[\xi_m^2] = \sigma_m^2 \le \rho^m \sigma_0^2$

$$P\Big(\sup_{m\ge k} |X_m - X_k| > \varepsilon\Big) \le \frac{1}{\varepsilon^2} \sum_{m=k}^{\infty} \sigma_m^2 \;\le\; \frac{\sigma_0^2}{\varepsilon^2} \sum_{m=k}^{\infty} \rho^m \;=\; \frac{\sigma_0^2}{\varepsilon^2} \, \frac{\rho^k}{1-\rho}.$$

D_k $D_k = R_k + X_k$ $R_k \le \rho_f^k$ $r_k := f(\hat{\eta}_{k-1}) - f(\eta^*)$ $|r_k| \le \rho_f |D_{k-1}|$ $|R_k| \le \rho_f^k |D_0|$

$\varepsilon := \sigma_0 \rho^{k/2} \sqrt{\log k}$ $\rho \le 1/2$ $\rho^{k/2}$

$$P(|D_k| > 2\varepsilon) \le P(|R_k| > \varepsilon) + P(|X_k| > \varepsilon)$$
$$\le \mathbf{1}_{[|R_k|>\varepsilon]} + \frac{\sigma_0^2}{\varepsilon^2} \frac{\rho^k}{1-\rho}.$$

$|R_k| \le \rho_f^k |D_0| \le \rho^k$ $\rho_f \le \rho$ k $\rho^k < \varepsilon = \rho^{k/2}\sqrt{\log k}$ $\rho^{k/2} < \sqrt{\log k}$ $k \ge 1$ $\mathbf{1}_{[|R_k|>\varepsilon]} = 0$ k

$$P\Big(|D_k| > 2\sigma_0 \rho^{k/2} \sqrt{\log k}\Big) \le \frac{\sigma_0^2}{\sigma_0^2\, \rho^k \log k} \frac{\rho^k}{1-\rho} = \frac{1}{(1-\rho)\log k}.$$

$k \to \infty$ $\sum_{k=1}^{\infty} 1/((1-\rho)\log k) = \infty$ Borel–Cantelli

$k_n = \lfloor n^2 \rfloor$

$$\sum_{n=1}^{\infty} P\Big(|D_{k_n}| > 2\sigma_0 \rho^{k_n/2} \sqrt{\log k_n}\Big) \le \sum_{n=1}^{\infty} \frac{1}{(1-\rho)\log(n^2)} = \frac{1}{2(1-\rho)} \sum_{n=1}^{\infty} \frac{1}{\log n} < \infty?$$

$\sum_{n=2}^{\infty} 1/\log n = \infty \int_2^{\infty} dx/\log x = \infty$ Borel–Cantelli

$\rho \le 1/2$ Chebyshev Borel–Cantelli

$n_k = \lfloor \alpha^k \rfloor$ $\alpha > 1$

$$\sum_{k=1}^{\infty} P(|D_{n_k}| > \varepsilon_{n_k}) \le \sum_{k=1}^{\infty} \frac{\sigma_0^2}{1-\rho} \frac{\rho^{n_k}}{\varepsilon_{n_k}^2}.$$

$\varepsilon_{n_k} = \sigma_0 \rho^{n_k/4}$ $n_k \to \infty$ $\rho^{n_k/4}$

$$\sum_{k=1}^{\infty} \frac{\rho^{n_k}}{\rho^{n_k/2}} = \sum_{k=1}^{\infty} \rho^{n_k/2} \le \sum_{k=1}^{\infty} \rho^{\alpha^k/2}.$$

$\rho \le 1/2 \rho^{\alpha^k/2} \le 2^{-\alpha^k/2}$ $\sum 2^{-\alpha^k/2}$ α^k k Borel–Cantelli $|D_{n_k}| \le \varepsilon_{n_k}$ k a.s.

n k $n_k \le n < n_{k+1} |D_n| \le \rho^{n-n_k} |D_{n_k}| +$ $|D_n| \le C\rho^{n/2}\sqrt{\log n}$ a.s.

?? (i) (ii) □ □

Remark 3.6 (/ Honest Critique of Theorem RA.1)**.** *[Honest Critique]*

1 $\rho < 1$ ρ $\mathbb{V}[\hat{\eta}_k \mid \eta^*]$ η^* ρ $\epsilon_{k+1}/\epsilon_k$ ρ *proxy*

2 ***Doob*** L_1 $|D_k| \le 1$ *a.s.* $\hat{\eta}_k \in [0,1]$ L_1 $[0,1]$ *SCX* $\hat{\eta}_k$ $[0,1]$ L_1

3 *(ii)* $\rho \le 1/2$ *Borel–Cantelli* C $\sqrt{\log k}$ *Fuk–Nagaev*

4 M_k σ_0 M_k σ_0 σ_0

RA.1 [Rigorous][Open]

3.2 Theorem RA.2

Theorem 3.7 (/ Infinite Regress Inevitability)**.**

Formal Statement

Assumption 3.8 (/ No Decay)**.** $\rho \ge 1$ $\sigma_{k+1}^2 \ge \rho\,\sigma_k^2$ $\rho \ge 1$

Assumption 3.9 (/ IID Increments)**.** $\rho = 1$ $\{\xi_k\}_{k=0}^{\infty}$ *i.i.d.* $\sigma^2 > 0$ $\xi_k \sim N(0, \sigma^2)$

Assumption 3.10 (/ Trivial Operator)**.** f $\rho = 1$ $|f(\eta) - f(\eta')| \approx |\eta - \eta'|$ $f(\eta) \equiv \eta$

$$\hat{\eta}_{k+1} = \hat{\eta}_k + \xi_k.$$

f $\rho \ge 1$

Theorem 3.11 (Infinite Regress under $\rho \ge 1$)**.** **??**–**??**

(i) $\rho = 1$ $\{\hat{\eta}_k\}$ *Pólya* ***recurrent*** *1*

$$\limsup_{k\to\infty} |\hat{\eta}_k| = \infty \quad a.s.,$$

$\varepsilon > 0 \lim_{k\to\infty} P(|\hat{\eta}_k - \eta^*| < \varepsilon) = 0$

(ii) $\rho > 1$ σ_k^2 $\sigma_k^2 \ge \rho^k \sigma_0^2$

$$\mathbb{E}[(\hat{\eta}_k - \eta^*)^2] \;\ge\; \sigma_0^2\,\rho^k \;\to\; \infty.$$

(iii) *recurrenceconvergence*

. $\rho = 1$ $\rho > 1$

I$\rho = 1$
?? $\hat{\eta}_{k+1} = \hat{\eta}_k + \xi_k$ ξ_ki.i.d. $\sim (0, \sigma^2)$ $S_0 := \hat{\eta}_0 - \eta^*$

$$\hat{\eta}_k - \eta^* \;=\; S_0 + \sum_{j=0}^{k-1} \xi_j.$$

1 **Pólya** Pólya (1921)

3.2 (Pólya). $\{X_n\}_{n=1}^{\infty}$ $\mathbb{R}$ *i.i.d.* $\mathbb{E}[X_1] = 0$ $0 < \mathbb{E}[X_1^2] < \infty$ $S_n := \sum_{i=1}^n X_i S_0 := 0$

1. $\limsup_{n\to\infty} S_n = \infty$ *a.s.* $\liminf_{n\to\infty} S_n = -\infty$ *a.s.*
2. $P(S_n = 0$ *i.o.*$) = 1$ *"i.o."*
3. $\varepsilon > 0 P(|S_n| < \varepsilon$ *i.o.*$) = 1$
4. S_n $\lim_{n\to\infty} S_n$

?? . (i) Kolmogorov $\limsup S_n$ $\pm\infty S_n/\sqrt{n} \xrightarrow{d} N(0,\sigma^2)$ $P(\limsup S_n = \infty) = 1$ $\liminf S_n = -\infty$ a.s.

(ii) Pólya Stirling $P(S_{2n} = 0) \sim C/\sqrt{n}$ $\sum P(S_{2n} = 0) = \infty$ Borel–Cantelli

(iii) (ii) S_n $\varepsilon > 0$ $P(|S_n| < \varepsilon$ i.o.$) = 1$

(iv) S_n L $S_n - S_{n-1} \to 0$ $S_n - S_{n-1} = X_n$ 0 $\mathbb{E}[X_n^2] = \sigma^2 > 0$ $\pm\infty$ $\liminf \neq \limsup$ (i) □

2

$X_j := \xi_j S_k := \sum_{j=0}^{k-1} \xi_j$ **??**X_ji.i.d. $\mathbb{E}[X_j] = 0 \mathbb{V}[X_j] = \sigma^2 < \infty$ **??**

$$\limsup_{k\to\infty} |S_k| = \infty \quad \text{a.s.}$$

$\hat{\eta}_k - \eta^* = (\hat{\eta}_0 - \eta^*) + S_k$ $\limsup_{k\to\infty} |\hat{\eta}_k - \eta^*| = \infty$ a.s.

3 $\varepsilon > 0$ k S_k

$$P(|\hat{\eta}_k - \eta^*| < \varepsilon) = P\big(|(\hat{\eta}_0 - \eta^*) + S_k| < \varepsilon\big).$$

$k \to \infty S_k/\sqrt{k} \xrightarrow{d} N(0,\sigma^2)$ S_k Chebyshev

$$P(|\hat{\eta}_k - \eta^*| < \varepsilon) \ \leq \ \frac{\mathbb{V}[\hat{\eta}_k - \eta^*]}{\varepsilon^2} = \frac{k\sigma^2}{\varepsilon^2} \to \infty?$$

Chebyshev 0 $(\hat{\eta}_k - \eta^*)/\sqrt{k\sigma^2} \xrightarrow{d} N(0,1)$

$$P(|\hat{\eta}_k - \eta^*| < \varepsilon) \approx P(|N(0,k\sigma^2)| < \varepsilon) = 2\Phi(\varepsilon/(\sigma\sqrt{k})) - 1 \to 0.$$

$k \to \infty \Phi(\varepsilon/(\sigma\sqrt{k})) \to \Phi(0) = 1/2$ $2\Phi - 1 \to 0$ $\lim_{k\to\infty} P(|\hat{\eta}_k - \eta^*| < \varepsilon) = 0$

4

- **Recurrence** $P(|\hat{\eta}_k - \eta^*| < \varepsilon$ i.o.$) = 1$ 1 η^* ε-
- **Convergence** $\lim_{k\to\infty} \hat{\eta}_k = \eta^*$ a.s. η^*

II$\rho > 1$
$\sigma_{k+1}^2 \geq \rho\,\sigma_k^2$ $\rho > 1$ $\sigma_k^2 \geq \sigma_0^2\,\rho^k$
$D_k = \hat{\eta}_k - \eta^*$

$$\mathbb{E}[D_{k+1}^2] = \mathbb{E}[(f(\hat{\eta}_k) - f(\eta^*) + \xi_k)^2] \geq \mathbb{E}[\xi_k^2] = \sigma_k^2 \geq \sigma_0^2\,\rho^k,$$

f ξ_k D_k f $\rho_f < 1$ $\rho_\sigma > 1$
$\lim_{k\to\infty} \mathbb{E}[D_k^2] = \infty$ Markov $M > 0$

$$P(|D_k| > M) \geq 1 - \frac{\mathbb{E}[D_k^2]}{M^2} \to 1 \quad k \to \infty \quad \mathbb{E}[D_k^2] \to \infty.$$

D_k $|D_k|$ 1
□ □

Remark 3.12 (/ Honest Critique of Theorem RA.2)**.** *[Honest Critique]*
RA.2

- ***1 Pólya i.i.d.*** *ξ_k* ***i.i.d.*** *Pólya $\{\xi_k\}$ $\mathcal{A}_{k+1}$ $\mathcal{A}_k$ ξ_k ξ_{k+1} ξ_k $\rho = 1$ i.i.d.*
- ***2*** *$\rho = 1$ f f $\rho_f < 1$ f f $\rho = 1$*
- ***3*** *$P(|\hat{\eta}_k - \eta^*| < \varepsilon) \to 0$ CLT k Berry–Esseen $O(1/\sqrt{k})$ " 0"*
- ***4*** *$\rho = 1$ $\epsilon_k \approx \epsilon_{k+1}$ prohibitive*

RA.2 i.i.d.[Rigorous]ξ_k i.i.d.[Conditionally Rigorous]

3.3 Conjecture RA.1

Conjecture 3.13 (/ Fixed-Point–Audit Heat Death Equivalence)**.** η^* *AE-* $H_{\min} = H(\varepsilon \mid S)$

$$\eta^* \;=\; H_{\min} \;=\; H(\varepsilon \mid S). \tag{5}$$

3 irreducible ignorance

Supporting Evidence

(i) $f(\eta) = a\eta + b$ $0 \leq a < 1$$b \in [0, 1-a]$ $f : [0,1] \to [0,1]$ $\eta^* = a\eta^* + b$

$$\eta^* = \frac{b}{1-a}.$$

S $H(\varepsilon \mid S)$ SCX Cercis $\varepsilon \sim \text{Bernoulli}(\theta)$ S $\text{Beta}(\alpha, \beta)$

$$H_{\min} = -\hat{\theta}\log\hat{\theta} - (1-\hat{\theta})\log(1-\hat{\theta}),$$

$\hat{\theta} = (\alpha + S)/(\alpha + \beta + n)$ (a, b) $b/(1-a)$

(ii) η^* f $\eta^* = f(\eta^*)$ $H_{\min} = H(\varepsilon \mid S)$ SCX $\mathcal{I}_A^{(\infty)}$ $H(\varepsilon \mid S)$ “”

(iii) / SCX η^* $H_{\min}$ nats bits - $\eta^* = h^{-1}(H_{\min})$ $h(p) = -p \log p - (1 - p)\log(1-p)$

(iv) RA.1 $\rho < 1$ η^* AE-$H_{\min}$ SCX

Remark 3.14 (/ Status). *[Open]*
Conjecture **??** *f SCX σ--audit-information projection operator*

$$f(\eta) = \mathbb{E}[\varepsilon \mid \mathcal{I}_A,\ \hat{\eta} = \eta],$$

$H(\varepsilon \mid S)$

Remark 3.15 (/ Honest Critique of Conjecture RA.1). *[Honest Critique]*

1. *(i) SCX f $\eta^* = b/(1-a)$ $H_{\min}$*
2. *(ii) $\mathcal{I}_A^{(\infty)}$ f functoriality*
3. *h (iii) h^{-1} $[0,1]$ $h(p) = h(1-p)$ $H_{\min}$ η^* η^**
4. *η^* η^* $\mathcal{A}_0$ $H_{\min}$ SCX η^* f*

3.4 Gödel

Open Problem 3.16 (SCX Gödel Correspondence / SCX Gödel).

$$\text{“RA- } \rho \geq 1\text{”} \longleftrightarrow \text{“PA ”}, \tag{6}$$

absolutely unauditable audit proposition *SCX* *[Open]*

Remark 3.17 (Gödel / Limitations of the Gödel Analogy). *[Honest Critique]*
Gödel

1. *Gödel RA- “” “”*
2. *Gödel RA-…syntacticstatistical*
3. *$\rho < 1$ $\rho \geq 1$ Gödel $\rho < 1$ “” Gödel $\rho < 1$*
4. *SCX Gödel “ ε ” SCX*

Gödel “”probabilistic logicaudit calculus *[Open]*

4

4.1

ρ	
$\rho < 1/2$	[Rigorous]
$1/2 \leq \rho < 1$	[Rigorous]
$\rho = 1$	[Rigorous]
$\rho > 1$	[Rigorous]

4.2

- **Theorem RA.1** () [Rigorous] **??–??** $\rho \leq 1/2$
- **Theorem RA.2** () [Rigorous] i.i.d. i.i.d.
- **Conjecture RA.1** () [Open]
- **Gödel** [Open]

4.3

$\rho < 1$

(i) $\mathcal{A}_0 =$ AI $\sigma\mathcal{A}_1 =$ $\sigma\mathcal{A}_2 =$ σ $\rho < 1$

(ii)

(iii)

$\rho \approx 1$ SCX

4.4

- **TS-** TS Sprint RA- ρ v_{drift} σ_k^2
- **AE-** **??** η^* $H_{\min}$
- **HC-** $\rho \ll 1$
- **FA-** $\prod_k \rho_k < 1$

5 Conclusion

RA-Theorem

(1) $\rho < 1$ [Rigorous]

(2) $\rho \geq 1$ [Rigorous]

(3) 3 $H_{\min}$ [Open]

(4) Gödel [Open]

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